

increase by the year? Cookies are becoming more ubiquitous. News sites offer personalised news. When you make a purchase at Amazon, it makes a list of 'similar items'—items bought by others who made the same purchase. The recommendations aren't based on only that list of similar items, but also on what you've bought before.

In an interesting twist, even spammers deliver spam based on the last few messages you opened. Advertisements are delivered to your browser, based on what ads you clicked on before. Bayesian techniques also promise to infuse search with an unheard-of degree of personalisation, since there's a whole lot of data that the user enters—search queries that patterns can be derived from. Microsoft and Google are already looking at Bayesian search.

From Bayes' theorem to BBNs

To see how BL works, an understanding of Bayes' theorem is essential. Suppose we know the probabilities of three things—that of catching a cold (A), that of catching a fever (B) and that of catching a fever if you already have a cold (C). Bayes' theorem says that the probability of someone with a fever having a cold is simply C times A, divided by B. Things such as A, B and C are called events; and in BBNs, events are represented as nodes in a graph, with their associated probabilities.

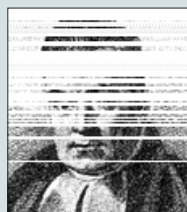
Data about events—their probabilities, whether they took place, etc—are 'pumped into' the nodes. When data comes in at one node, it is sent to all other connected nodes. The network is updated each time a node receives fresh data.

Let's add two variables to the fever and cold example—flu and pneumonia. A BBN

is typically drawn such that if a node is beneath another and connected to it, it is dependent on it—for example, a cold can cause a fever, so the 'fever' node would be above, connected to the 'cold' node beneath. Now, for example, say we know that pneumonia causes fever, and that flu does not. If the patient contracts a fever, data gets pumped into the 'fever' node. The 'pneumonia' node is affected by this, gets updated, and the 'belief' of the network is set to new probability values—the 'pneumonia' node would now have a higher probability associated with it. On the other hand, the 'flu' node would not be affected. Dependencies can also be

The Rev. Thomas Bayes

An English theologian, Thomas Bayes was a Presbyterian minister who happened to be a very good mathematician. He argued that the existence of God could be deduced from equations. He wrote an essay on his theory of probability, which was published by the Royal Society in 1764. He also set out a method of deducing the future from past events.



His findings and methods have always been controversial. And Bayes' theorem, as such, was not published until after his death.

deduced by the system: if, often, the 'flu' node is often sent a value of '1' just after

Read My Lips

Researchers have found that adding lip reading to conventional speech recognition (SR) makes for better accuracy, especially in noisy spaces. And Intel has come up with software—Audio Visual Speech Recognition (AVSR)—which, combined with Intel's OpenCV computer vision library, lets SR programmers build lip reading into their software. The AVSR combined with OpenCV can detect one face amongst several, and observe that person's mouth movements.

the 'fever' node is sent a value of '1', the system may deduce that fever causes flu.

Bayesian Learning is essentially the process of the network changing its values, based on the values that are pumped into certain nodes. Each and every causal relationship need not be specified in a BBN: the network can deduce likely dependencies on its own.

A system called PicHunter gives a flavour for how Bayes' theorem can work in real life. PicHunter finds a picture based on certain parameters: a picture that comes under a certain category, or a picture that looks like another, etc. Now, say you want to find a picture that looks like one of a certain dog. You look at some pictures, and say 'hot' or 'cold', based on whether the picture shown to you is similar or not to the one you're looking for. A non-Bayesian system might analyse the pictures you choose, and try and find the one you're looking for based only on that. PicHunter finds a pattern in the sequence of pictures you've been choosing, and builds your profile. The problem is changed from that of finding the picture to the problem of building a user profile! Once this user profile has been built, it's a matter of comparing it to PicHunter's model of what users would do if they wanted a particular picture, and thus delivering the correct picture.

Damn that spam!

Spam filtering is hot right now. It won't be, if Mr Gates' proposal of ISPs charging a penny per e-mail sent is implemented, but that doesn't seem likely—and so we have a whole bunch of commercial and open-source spam blockers, each claiming to be the best. Several of these use BL as part of their repertoire of methods. Anti-spamming is ideally suited to BL, because it



Computers To Mature With Experience

Intel's new Probabilistic Networks Library (PNL) and Open Source Machine Learning Library (OSML) are collections of software using which computers might be able to learn from experience. OSML is free, and is available from Intel's Web site. Machine Learning (ML) is the discipline dealing with, of course, machines learning, and thus being able to make predictions, among other things. And BL is a methodology that's being increasingly embraced by ML folks.

OSML is basically a toolkit, which incorporates Bayesian Learning methods, for designers of computers

and robots. The PNL and OSML can also be used in enhancing applications, for example, gene-database searching, and, of course, spam classifying.

Using typical Bayesian phraseology, David Tennenhouse, Intel's Director of Research, says, "Computers need to be able to learn from their experiences with users, and the world around them. Using new statistical methods to identify key patterns, these systems will start anticipating the needs of their users and pre-computing responses to the most likely questions, so that the answers will be instantly available the moment they are needed."